Three Interpretations of the Denoising Process

The Denoising Matching Term $\mathcal{D}_{KL}\left(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\right)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)=\frac{q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$ Baye's Rule

Pamaterisation of the conditional distributions

Time dependent noise α_t and $\bar{\alpha}_t=\prod_{i=1}^{t}\alpha_i$

$q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)=q(\mathbf{x}_t|\mathbf{x}_{t-1})=\mathcal{N}(\mathbf{x}_t;\sqrt{\alpha_t}\mathbf{x}_{t-1},(1-\alpha_t)\mathcal{I})$

$q(\mathbf{x}_t|\mathbf{x}_0)=\mathcal{N}(\mathbf{x}_t;\sqrt{\bar{\alpha}_t}\mathbf{x}_0,(1-\bar{\alpha}_t\mathbf{I})$

$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)=\mathcal{N}\left(\mathbf{x}_{t-1};\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t+\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_t},\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{I}\right)$

$\boldsymbol{\mu}_q(\mathbf{x}_t,\mathbf{x}_0)$ $\quad$ $\boldsymbol{\Sigma}_q(t)=\sigma_q^2(t)$

Parameterise the denoising process also as a Gaussian, align with forward process

$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)=\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_q,\sigma_q^2)$

$p(\mathbf{x}_{t-1}|\mathbf{x}_t)=\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_\theta,\sigma_q^2)$

$$\underset{\theta}{\operatorname{argmin}}\,\mathcal{D}_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))=\underset{\theta}{\operatorname{argmin}}\,\frac{1}{2\sigma_q^2(t)}\left\|\boldsymbol{\mu}_\theta-\boldsymbol{\mu}_q\right\|_2^2$$

Now rewrite the mean in terms of the samples

$$\boldsymbol{\mu}_q(\mathbf{x}_t,\mathbf{x}_0)=\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t+\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_t}$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t,\mathbf{x}_0)=\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t+\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_\theta(\mathbf{x}_t,t)}{1-\bar{\alpha}_t}$$

First Interpretation: Denoising process aims to reconstruct the original sample from the noised data

$$\underset{\theta}{\operatorname{argmin}}\,\mathcal{D}_{KL}=\underset{\theta}{\operatorname{argmin}}\,\frac{1}{2\sigma_q^2(t)}\frac{\bar{\alpha}_{t-1}(1-\alpha_t)^2}{(1-\bar{\alpha}_t)^2}\left\|\mathbf{x}_\theta(\mathbf{x}_t,t)-\mathbf{x}_0\right\|_2^2$$

from one-shot sampling $\mathbf{x}_t=\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon}_0$, we have $\mathbf{x}_0=\frac{\mathbf{x}_t-\sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon}_0}{\bar{\alpha}_t}$

$$\boldsymbol{\mu}_q(\mathbf{x}_t,\mathbf{x}_0)=\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\boldsymbol{\varepsilon}_0$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_\theta,\mathbf{x}_0)=\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\boldsymbol{\varepsilon}_\theta(\mathbf{x}_t,t)$$

Second Interpretation: Denoising process requires a model to learn the noise added in the forward process, and removes it

$$\underset{\theta}{\operatorname{argmin}}\,\mathcal{D}_{KL}=\underset{\theta}{\operatorname{argmin}}\,\frac{1}{2\sigma_q^2(t)}\frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\alpha_t}\left\|\boldsymbol{\varepsilon}_\theta(\mathbf{x}_t,t)-\boldsymbol{\varepsilon}_0\right\|_2^2$$

From the Tweedie's formula, for a Gaussian variable $z\sim\mathcal{N}(z;\mu_z,\sigma_z)$, we have the expectation value for the mean $\mathbb{E}[\mu_z|z]=z+\sum_z\nabla_z\log p(z)$.

$\mathbb{E}[\boldsymbol{\mu}_{\mathbf{x}_t}|\mathbf{x}_t]=\mathbf{x}_t+(1-\bar{\alpha}_t)\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)=\sqrt{\bar{\alpha}_t}\mathbf{x}_0$

$$\mathbf{x}_0=\frac{\mathbf{x}_t+(1-\bar{\alpha}_t)\nabla\log p(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}$$

$$\boldsymbol{\mu}_q(\mathbf{x}_t,\mathbf{x}_0)=\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t,\mathbf{x}_0)=\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{\alpha_t}}s_\theta(\mathbf{x}_t,t)$$

Third Interpretation: (**Score-Matching**) Learning the score of the data $\nabla p(x)$ at an arbitrary noise level

$$\underset{\theta}{\operatorname{argmin}}\,\mathcal{D}_{\mathrm{KL}}=\underset{\theta}{\operatorname{argmin}}\,\frac{1}{2\sigma_q^2(t)}\frac{(1-\alpha_t)^2}{\alpha_t}\left\|s_\theta(\mathbf{x}_t,t)-\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)\right\|_2^2$$